

SE Sem-IV

MDM (Fundamentals of Telecommunication Systems)

IA-2 Question Bank

Module 4

- Draw and explain the block diagram of telephone system. Also, list its functions.
- Draw and explain the block diagram of mobile system.
- Draw and explain the block diagram of modem. Also, explain its working principle.
- What are telecommunication standards? Briefly discuss the need for telecommunication standards.
- Give a comparative summary of ITU, IEEE, and 3GPP in tabular format.

Module 5

- **Numerical** on AM, FM and PM.
- Give a comparative summary of AM, FM and PM.
- Discuss frequency division multiplexing (FDM). Explain multiplexing and demultiplexing process as applied to FDM with the required diagrams.
- Give advantages, disadvantages and applications of FDM.
- Discuss time division multiplexing (TDM). Draw the required diagrams.
- Give advantages, disadvantages and applications of TDM.
- With the help of an example explain the process of Hamming code generation and its usage for error detection and correction.
- **Numerical** on Hamming code, simple parity, CRC and checksum methods.

Module 6

- What is IoT? State its characteristics.
- Draw and explain the layered architecture of IoT.
- Discuss the features of 5G.
- What is VoIP? Explain its working.
- What is cloud communication? Explain its core components.
- Give a comparative study of 5G, VoIP and cloud communication.